

MDS-UPDRS Finger Tapping Score

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested.

Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [222, 192, 223, 159, 167, 219, 141, 189, 155, 224]

Tap amplitudes (pixels): [178.21896644296868, 168.9644933114647, 174.41330224498358, 159.84992962150469, 170.66048165876012, 168.58232410309213, 160.5615146914104, 152.620444239951, 162.78820596099706, 158.14234094637652, 154.38264151127873]

GPT-5

MDS-UPDRS Item 3.4 (Finger Tapping) Scoring Analysis

Interpreting your features for 10 consecutive taps (speed via periods; amplitude in pixels). You have 11 amplitude values and 10 periods, with “Number of taps: 11,” which typically implies 11 detected peaks (10 intervals). The scoring, however, is based on the quality of a 10-tap sequence. I will assess:

- Speed (slowing)
- Rhythm (interruptions/hesitations/arrests)
- Amplitude trend (decrement timing)

1) Rhythm interruptions/hesitations

- Tap periods (frames): [222, 192, 223, 159, 167, 219, 141, 189, 155, 224]
- These vary but remain within a reasonable band without extreme outliers or pauses. No clear evidence of >5 interruptions or a prolonged arrest.
- Some variability suggests possible slight hesitations, but it does not convincingly reach 3–5 distinct interruptions.

Assessment: At most slight irregularity; not consistent with “3–5 interruptions” or “>5 interruptions/long arrest.”

2) Speed (slowing)

- Periods fluctuate between 141–224 frames with no monotonic slowing trend. Variability is present but does not indicate sustained mild or moderate slowing across the sequence.

Assessment: Slight slowing at most.

3) Amplitude decrement

- Amplitudes: [178.2, 169.0, 174.4, 159.8, 170.7, 168.6, 160.6, 152.6, 162.8, 158.1, 154.4]
- Relative to the first tap (178.2), amplitudes are generally lower thereafter and fluctuate, with a tendency to be below the first from early taps onward.
- Key for scoring is when the decrement clearly sets in:
 - Early taps: Tap 2 (169) and Tap 3 (174) are already below the first, and by Tap 4 it's down to ~159.8.
 - The sequence does not show a “near-end only” decline; the reduction is evident from early/mid taps onward, and remains lower with some oscillation.

Assessment: The amplitude decrement begins early (after the 1st–2nd tap), closer to “starting after the 1st tap” than to “midway” or “near the end.”

Overall Mapping to Criteria

- Interruptions: Not clearly in the 3–5 range; no long arrest. Points away from scores 2–3 based on interruptions alone.
- Speed: At most slight slowing (supports score 1).
- Amplitude decrement timing: Begins early (after the first tap), which maps to score 3 (Moderate) criterion “The amplitude decrements starting after the 1st tap.”

Given MDS-UPDRS scoring uses the worst applicable criterion, the early amplitude decrement aligns with a score of 3 even if interruptions/slowing are milder.

Final score: 3
